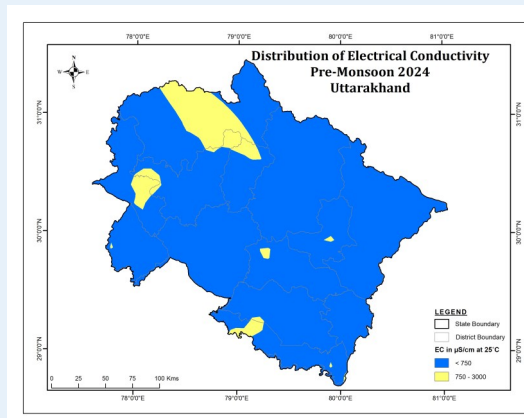


Groundwater Quality Scenario in Uttarakhand

Parameters	No of samples	Permissible limit	No of Samples above permissible limit	% Samples above permissible Limit
EC	248	3000 $\mu\text{S/cm}$	0	0
Fluoride	248	1.5 mg/L	3	1.21
Nitrate	248	45 mg/L	33	13.31
Arsenic	104	10 ppb	0	0.00
Uranium	103	30 ppb	0	0.00



Districts with anomalous values at sporadic locations

EC (3000 $\mu\text{S/cm}$)	Not Any
Fluoride ($F > 1.5 \text{ mg/L}$)	Almora, Uttarkashi
Nitrate (Nitrate $> 45 \text{ mg/L}$)	Bageshwar, Champawat, Dehradun, Haridwar, Nainital, Pithoragarh, Us Nagar, Uttarkashi
Arsenic ($As > 10 \text{ ppb}$)	Not Any
Uranium ($U > 30 \text{ ppb}$)	Not Any

For Further Information, Contact to :

Chairman, CGWB, Bhujal Bhawan,
NH IV Faridabad, Haryana - 121001



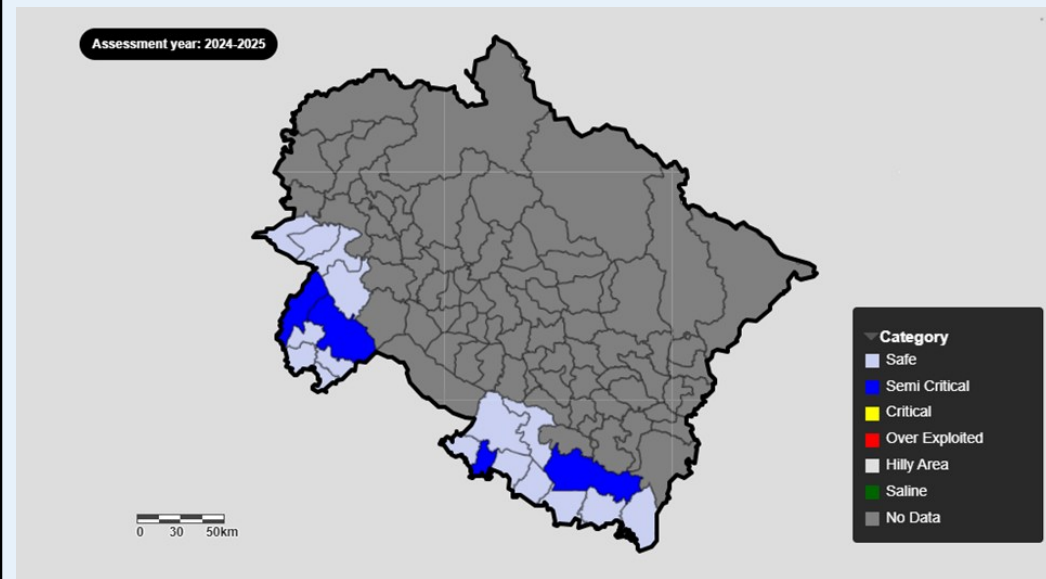
<https://inqres.iith.ac.in>

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<https://www.facebook.com/cgwb.chg>



Central Ground Water Board
Department of Water Resources, RD & GR
Ministry of Jal Shakti, Government of India



**Dynamic Ground Water Resources, Ground Water Level &
Ground Water Quality of Uttarakhand, 2025**

November, 2025

Background

- ◆ Ground Water Resources Assessment (GWRA)- jointly carried out by Central Ground Water Board and State Nodal/Ground Water Department periodically as per the Ground Water Resource Estimation Committee (GEC) methodology.
- ◆ Carried out under the guidance of the respective State/UT Level Committees (SLCs) and overall supervision of Central Level Expert Group (CLEG).
- ◆ As part of the assessment, 'Annual Extractable Ground Water Resource' as well as 'Annual Ground Water Extraction' are assessed for each assessment unit (Block).
- ◆ The 'Stage of Ground Water Extraction' is computed as the ratio of 'Annual Ground Water Extraction' with respect to 'Annual Extractable Ground Water Resource' and is usually expressed in percentage. Based on the stage of extraction, the assessment units are categorized as Safe ($\leq 70\%$), Semi-Critical ($>70\%$ and $\leq 90\%$), Critical ($>90\%$ and $\leq 100\%$) and Over-Exploited ($>100\%$).
- ◆ GWRA- 2025, 2024, 2023, 2022 and 2020 has been carried out through a software/web-based application "INDIA-GROUNDWATER RESOURCE ESTIMATION SYSTEM (IN-GRES)" developed by CGWB through IIT-Hyderabad.

Salient Features

1	Rainfall	1,537.59 mm
2	Hydrogeology	About 85 % of the State is mountainous and underlain by hard rock. Around 15 % of the geographical area is underlain by semi-consolidated and unconsolidated formations (Tarai and bhabar).
3	Recharge Worthy Area of the State	6.30 Thousand Sq. Km
4	Assessment Unit (AU) Type / Number	Block / 20 Numbers
5	Average area of Assessment Unit	314.93 Sq. Km

Findings

	Attribute	GWRA-2017	GWRA-2020	GWRA-2022	GWRA-2023	GWRA-2024	GWRA-2025
1	Total Annual Ground Water Recharge (in bcm)	3.04	2.02	2.01	2.02	2.14	2.13
2	Annual Extractable Ground Water Resources (in bcm)	2.89	1.85	1.86	1.85	1.96	1.95
3	Annual Ground Water Extrac-	1.64	0.87	0.89	0.95	1.05	1.05
4	Stage of Ground Water Extrac-tion (in %)	56.83	46.8	48.04	51.69	53.54	53.92

bcm: Billion Cubic Meters

Categorization of Assessment Units based on the 'Stage of Ground Water Extraction'

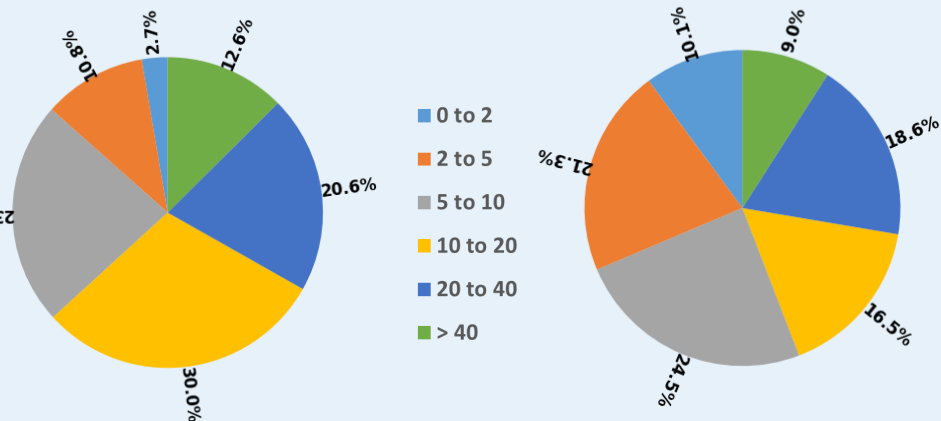
S. No	Category	GWRA-2017		GWRA-2020		GWRA-2022		GWRA-2023		GWRA-2024		GWRA-2025	
		Num-ber of AUs	% of AUs	Num-ber of AUs	% of AUs	Num-ber of AUs	% of AUs	Num-ber of AUs	% of AUs	Num-ber of AUs	% of AUs	Num-ber of AUs	% of AUs
1	Safe	13	72	14	78	14	78	14	77.78	16	80	16	80
2	Semi-critical	5	28	4	22	4	22	4	22.22	4	20	4	20
3	Critical												
4	Over-exploited												
5	Saline												
Total number of AUs		18		18		18		18		20		20	

Recommendations: -

- * Uttarakhand State has a distinct geological attribute with wide variety of rock units ranging in age from Archean to Quaternary. About 85 % of the geographical area of the state is mountainous and underlain by hard rocks. Ground water in the hard rock area is harnessed through the springs and hand pump tapping the weathered zone. Discharge of springs in the Lesser Himalaya and Central Himalaya is variable and ranges from 60 to 600 lpm. About 15 % of the geographical area is underlain by semi-consolidated and unconsolidated formations known as Tarai and Bhabhar.
- * Total Annual Ground Water Recharge in the State (2025) has been assessed at 2.13 billion cubic meters (bcm). The Total Annual Extractable Ground Water Resource of the State has been assessed at 1.95 bcm. The Annual Ground Water Extraction of the State has been assessed as 1.05 bcm. The Stage of ground water extraction for the entire State has been estimated as 53.92 %.
- * Out of the 20 assessment units (blocks), 4 units (20%) have been categorized as 'Semi-critical' and 16 units (80 %) as 'Safe'. There is no saline block in the State.
- * In the safe category areas of Uttarakhand, State Government can judiciously develop the ground water resource mainly for agricultural use, however, at no point of time the extraction level should exceed 70%.
- * Development of Springs and their catchment in hilly areas.
- * National Aquifer Mapping & Management Programme (NAQUIM) Reports prepared by CGWB (<https://cgwb.gov.in/cgwbpnmi/>) which are also being shared with State/District Authorities and Ground Water Year Book published by CGWB having water level & water quality data may be used in Ground water management. (<https://cgwb.gov.in/cgwbpnmi/>).
- * Regulation & control of Ground water Extraction: Ministry of Jal Shakti has issued the guidelines for control and regulations of ground water extraction vide notification dated 24.09.2020 which has further been amended in March 2023. Concerned departments may ensure implementations of the guidelines.

Groundwater Level Scenario in Uttarakhand

1. Depth to Groundwater Level

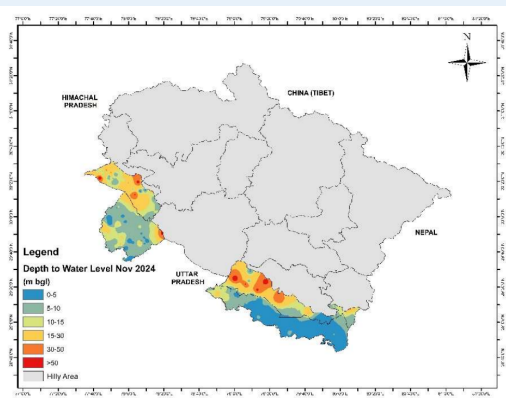
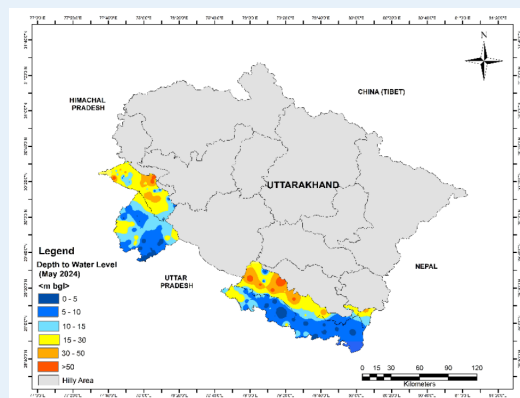


Pre-Monsoon (May 2024)

About 13.9% of wells recorded groundwater levels in 0–5 m bgl, 23.3% in 5–10 m bgl, 18.8% in 10–15 m bgl, and 43.9% in >15 m bgl, with overall levels ranging from 0.63 m to 96.82 m bgl, indicating predominantly deep groundwater in Uttarakhand with shallow pockets in some districts.

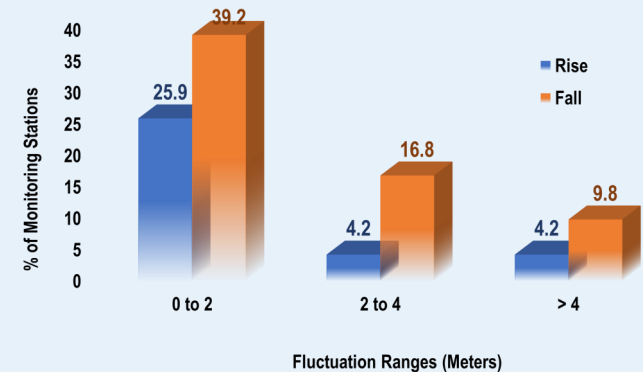
Post-Monsoon (November 2024)

About 27% of wells recorded groundwater levels in 0–5 m bgl, 26.1% in 5–10 m bgl, 10.6% in 10–15 m bgl, and 36.3% in >15 m bgl, with overall levels ranging from 0.03 m to 94.58 m bgl, indicating predominantly deep groundwater with shallow pockets in Udham Singh Nagar and Haridwar districts.



2. Decadal Fluctuation of Groundwater Levels

Pre-Monsoon (May 2024 vs Decadal Mean: May 2014–2023)



Post-Monsoon (November 2024 vs Decadal Mean: November 2014–2023)

